

Co-Attendance to Community:
Simulating Analog Social-Network Formation in Boutique Fitness

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A. Abstract

Repeated attendance in boutique fitness spaces has the potential to create meaningful social relationships, even in the absence of a formal digital social network. Nonetheless, it is relevant to examine how far analog co-attendance alone can realistically take this process. How does repeatedly attending the same SoulCycle studios, instructors, and class times translate into familiarity and stronger social ties between riders? Additionally, does coordination with existing social connections expand a rider's network, or does it instead reinforce already-established routines? This project addresses these questions by simulating a temporal SoulCycle rider network consisting of 10,000 synthetic riders over 52 weeks. Riders were assigned heterogeneous attendance rates, geographic constraints, and preferences for studios, instructors, days, and class slots, while repeated co-attendance gradually produced familiarity and active social ties. Across the simulation, analog co-attendance generated a coherent but extremely sparse, fragmented, and locally bounded network. Familiarity ties developed much more frequently than active social ties, while high-frequency riders accumulated greater familiarity degree because of their increased opportunities for repeated exposure. Greater coordination with existing social ties generally narrowed familiarity-network breadth by concentrating riders within established routines. These findings suggest that SoulCycle's analog environment is capable of generating meaningful community structure, but that many repeated encounters remain unrealized social opportunities. Lightweight digital features may therefore be most useful when they help riders recognize and connect with people they already encounter rather than replacing the analog experience with a conventional social network.

B. Introduction

Founded in 2006, SoulCycle has become one of the most recognizable companies in the boutique fitness industry. Its signature 45-minute indoor cycling classes have helped popularize the prevalence of boutique fitness and its associated communities worldwide. One characteristic that distinguishes SoulCycle from digitally integrated fitness companies such as Peloton and Orangetheory Fitness is its comparatively limited reliance on rider-facing technology to elevate the experience. Rather than connecting riders through leaderboards, digital profiles, or an explicit online social network, the SoulCycle experience remains largely centered on repeated in-person attendance at particular studios, with preferred instructors and class times. This distinction introduces a key question: how has SoulCycle's social network manifested solely from recurring analog co-attendance between riders?

Literature on this specific topic is limited, but research across several disciplines can help establish the broader context of fitness network formation. Yuan, Miao, and Liu examine digital infrastructures as network facilitators in *The R/Evolution of Civic Engagement: An Exploratory Network Analysis of the Facebook Groups of Occupy Chicago*. The researchers analyze the Facebook groups frequented by some random 600 members of Occupy Chicago, a 2010s political movement in opposition to concentrated corporate power. These participants belonged to 2,829 Facebook groups altogether, and surprisingly the subject matter of these groups extended far beyond politics. Over half of these groups revolved around non-political matters, with 28% percent organized around casual socialization, another 24% focused on cultural/artistic expression; only 10% were explicitly political (Yuan et al. 2019). These findings suggest that individuals initially connected via one shared interest may also participate in broader communities that extend beyond the original source of affiliation. However, the larger network

remained loosely connected, with a density of 0.02, and highly fragmented, with a modularity score of 0.84. The groups therefore formed distinct, internally connected clusters rather than one cohesive network. This pattern is relevant to boutique fitness because riders may similarly become connected through a shared activity while remaining organized into localized communities based on common studios, instructor preferences, and schedules.

A more anecdotal, though still relevant, account of digital infrastructure within fitness communities appears in Danielle Friedman's recent *New York Times* article, "Sick of Swiping, Some Daters Would Rather Sweat Together." Rather than analyzing fitness networks through a formal network analysis lens, Friedman documents the growing use of fitness as an environment for romantic connection. One example follows Nicole Ho and D.C. Banks, who met through Surf, a fitness-based dating app, and completed a Hyrox race together on their first date. Both D.C. and Nicole lauded the authenticity of centering a date around a physical activity: "At a fitness event like Hyrox, 'There's no faking it,' Mr. Banks said a few days after the race. 'You're in fight or flight, and you're just going with it.' Ms. Ho agreed. You can't hide when you're suffering, she said. 'You're huffing and puffing, you don't look cute,'" (Friedman 2026). Friedman also discusses a post-workout mixer held at SoulCycle's West Village studio, sponsored by another wellness-based dating app named Ateam, demonstrating that boutique fitness brands are now open to utilizing dating app infrastructures to intentionally facilitate customer-to-customer interaction. Friedman argues that these examples reflect a broader change in how younger adults socialize: as Gen Z spends more time and money on fitness and less on alcohol and nightlife, gyms and boutique studios may increasingly function as alternative social gathering spaces and dating locales. This suggests that shared exercise can serve as a tool with which fitness-focused individuals can meet others with similar routines, interests, and lifestyle

preferences. For this project, the article is most useful as evidence that digital tools can accelerate relationships that originate in fitness environments, raising the question of whether repeated in-person attendance with fellow athletes might produce similar communities without digital romantic infrastructures.

Existing literature also offers interpretations from business and management studies regarding how companies engineer communities within fitness spaces. In “Adding Linking Value to the Value Proposition,” a chapter from *Start-ups and the Mobilization of Social Interactions*, Barès, Cova and Nemani discuss how businesses (particularly SoulCycle) can incorporate what they call “linking value” into their products and services. Traditional value refers to the direct usefulness provided by the product itself. For example, a bakery such as Paris Baguette primarily attracts customers through the quality and palatability of its pastries. Linking value, however, describes the additional value a business creates by helping customers form relationships with one another. Under this framework, customers may patronize a business not only because they value the product being sold, but also because they value the community and shared experience surrounding it. Barès, Cova and Nemani identify SoulCycle as a particularly strong example of this model. They summarize it best by describing how the instructors are the architects of the company’s linking value: “In the words of a customer, ‘Usually at the beginning of class the instructor encourages you to introduce yourself to the people around you. This is your “team.”’ Throughout class the instructor will encourage the group by saying “let’s go team,” “ride for the person next you,” things of that nature,”’ (Bàres et al. 2023). These practices transform the class from a simple 45-minute cycling workout into a collective experience built around synchronicity, instructor motivation, and shared intensity. The studio environment and instructional philosophy therefore create connections not only between riders and instructors, but also among riders

themselves. Rather than treating community as a secondary marketing feature, SoulCycle incorporates it directly into the experience being purchased. This project takes that existing analog linking value as the starting point for simulating rider-network formation. It further considers whether digital mechanisms could extend relationships initiated inside the studio by helping riders recognize repeated co-attendees, coordinate future classes, and maintain connections beyond the duration of a single workout.

The final piece of literature directly addresses the relationship between network features and physical activity. This is especially important in the SoulCycle case because it suggests that a network formed through the company's linking value may also reinforce the functional value of its workout product. Prochnow and Patterson's article, "Assessing Social Network Influences on Adult Physical Activity Using Social Network Analysis: A Systematic Review," examines 28 studies of adult physical activity, with a unique focus on social network analysis. In doing so, the authors address two major research questions: how adult physical activity has been studied using social network analysis approaches and how an adult's social network is associated with physical activity behavior. Regarding the first question, they identify a significant methodological gap. Only four of the 28 studies used whole-network designs, while the remaining 24 focused on individual participants and their immediate personal networks. Many of these egocentric studies also restricted respondents to naming between three and 15 connections, most commonly five, and relied primarily on regression-based analyses (Prochnow and Patterson 2022). As a result, much of the existing literature examines how a small personal network relates to one individual's behavior rather than how physical activity operates across a complete network. This gap provides a useful justification for modeling SoulCycle as a whole network in which node-level, edge-level, and graph-level structure can be observed simultaneously.

The review's second research question offers stronger evidence that social network characteristics are associated with adult physical activity. One included study found that each additional contact in a participant's health network was associated with 10% greater odds of being physically active. Another found that regularly active adults had an average of 5.77 ties, compared with 4.86 among inactive adults. Regarding network composition, researchers discovered high levels of homophily: non-exercisers reported that 70% of their network contacts were also non-exercisers, while regular exercisers reported that 57% of their contacts exercised regularly. The review further identifies network exposure, the strength and reciprocity of connections, and the presence of exercise partners as recurring correlates of physical activity. Overall, the review suggests that network size, composition, connectivity, and exposure to active peers may shape an individual's likelihood of remaining physically active. This creates an important connection between SoulCycle's traditional and linking value. Efforts to facilitate rider relationships may not only strengthen belonging and customer community, but could also reinforce exercise consistency and sustained participation in the company's core product. In this sense, analog community formation and carefully designed digital mechanisms may produce value on both sides of the SoulCycle experience: stronger interpersonal connections and greater engagement with physical activity.

This project builds on the existing literature by simulating a temporal social network among SoulCycle riders using the limited available data on studios, instructors, attendance patterns, and rider behavior. The primary goal is to approximate the baseline level of connectivity that may emerge through repeated analog co-attendance and to consider how this existing network could be extended through digital mechanisms. The model incorporates several insights from the literature review, particularly SoulCycle's use of linking value and the

relationship between health-network characteristics and individual physical activity. By using a simulation rather than relying solely on ethnographic or egocentric accounts of rider behavior, the project can represent a much larger bounded network and examine its development over time. This makes it possible to derive node-level, edge-level, and graph-level statistics that describe the structure, fragmentation, clustering, and growth of the simulated fitness network. The results should therefore be interpreted not as a reconstruction of SoulCycle's actual rider network, but as a calibrated model of the social structure that its existing analog environment may be capable of producing.

C. Data Collection

Obtaining data for this project involved browsing a litany of publicly available data from SoulCycle's website. Data collection began with collecting features information on all of SoulCycle's current locations (excluding the brand's seasonal studios in the Hamptons). For each studio I collected the studio's ID abbreviation, name, state, and official region. Furthermore, I used the week of July 20th to the 26th as an observation week to gain an idea of a typical week's schedule for each studio. I collected the number of rides offered within the observation week per studio per location (the vast majority of locations contained only one studio), and for each studio I recorded the number of bikes available to be reserved. From this I was able to calculate the weekly slot offerings (number of classes in studio A x number of bikes in studio A + number of classes in studio B x number of bikes in studio B) as a proxy for scheduled capacity/potential attendance volume, with the reasonable assumption that SoulCycle factors in class attendance into each studio's schedules. Furthermore, I created subjective divisions of the given studios in order to easily create reasonable bounds on network formation, largely based on geography.

Namely, I created a network market, which places a studio in a market based on the major city/region that it resides in, and these did not always coincide with the official region listed on SoulCycle's website. For example, the four studios I placed in my DMV network market (14th, Bethesda, Clarendon, and Georgetown), are spread across three separate official regions on SoulCycle's website, whereas SoulCycle's Austin, Dallas, and Houston studios are all listed under one official Texas region online, whereas I consider them three separate markets. I also created a more fine-grained network cluster feature, largely to address the breadth of coverage SoulCycle has in the New York City area and to be realistic about actual rider/instructor behavior within the region; for example, in my model this network cluster variable makes it impossible for a rider to ride regularly at both SoulCycle's Greenwich, Connecticut and Hoboken, New Jersey locations, despite both being in my Greater NYC network market.

These weekly slot offerings were then aggregated by network market in order to approximate the relative scale of SoulCycle activity across regions. As shown in Figure 1, Greater NYC contains substantially more scheduled capacity than any other market, while the DMV, Southern California, and Northern California form a second tier of relatively large markets. Boston, Philadelphia, Chicago, and South Florida occupy a more moderate range, while Seattle, Dallas, Atlanta, Austin, Houston, and Ann Arbor have considerably smaller scheduled footprints. These differences were useful for calibrating the simulated environment because they prevented each market from being treated as if it contained the same level of rider activity. Importantly, these values represent scheduled reservable capacity rather than observed attendance, and should therefore be interpreted as a proxy for the relative scale of each market rather than direct ridership counts.

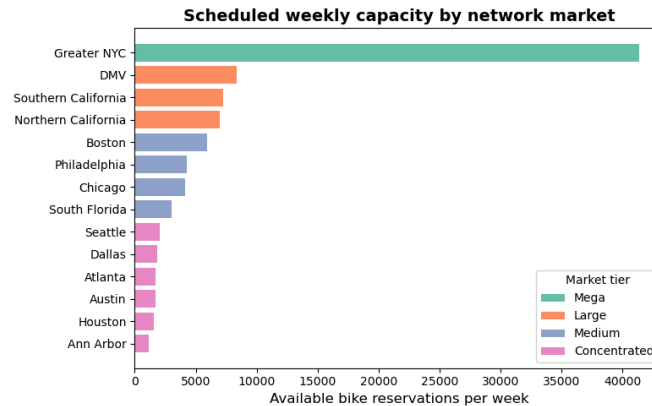


Figure 1: Weekly Bike Capacity by Market (SoulCycle 2026)

After establishing the studio environment, I transitioned into collecting instructor-level information in order to approximate how instructors move through and contribute to that environment. I first developed a spreadsheet of all active SoulCycle instructors listed on the company's regional instructor pages, monitoring for duplicates so that instructors teaching across multiple regions were not counted more than once. I also manually verified that instructors remained active by checking for upcoming classes through the end of August. This process resulted in 248 active instructors across the modeled markets. From this population, I constructed a stratified sample of 50 instructors in order to estimate typical instructor behavior while still maintaining representation across markets of substantially different sizes. For each sampled instructor, I recorded the number of classes taught during the July 20–26 observation week and the number of studios in which they taught. These observations were subsequently used to inform the instructor-generation and scheduling components of the simulation rather than reproducing the exact behavior of any individual instructor. Together, the studio and instructor data provided the observed company environment within which the synthetic rider population described in the following section was generated.

D. Model Construction

The largest underlying assumption made while developing this model was that rider behavior follows the “co-attendance to coordination” pipeline. This pipeline begins with two riders attending the same class; this is the “shared class” phase. These riders may encounter one another across several classes, or “repeated co-attendance”. The next step is when that repetition turns into recognition, or what we call a “familiarity” tie (for example, maybe you both have gone to the 7am class at a particular studio before work for the past three weeks). When these two continue to interact by encountering each other, their tie categorically changes from a “familiar” one to a stronger “social” one. Finally, once that social tie is achieved, we hit that final “coordination” stage where those with social ties actively foster relationships to the point where they coordinate going to certain classes together. To operationalize this process, the model assigns explicit class thresholds to each stage. Two riders form a familiarity tie after attending at least three classes together, and stronger social ties require at least six shared classes in addition to sufficient accumulated tie strength. These thresholds are not intended to represent universal cutoffs for when two people become familiar or socially connected, they are meant to provide a consistent rule for translating repeated exposure into progressively stronger network relationships.

Since repetition is the main driver of recognition and relationship building, the overall network is both temporal and weighted. Each shared class increases the tie strength of a pairwise relationship, while periods without interaction reduce that strength through weekly decay (a decay rate of 5%). Essentially, the model treats social connection as an evolving process rather than a static one. Once an active social tie has formed, behavioral influences change. Up to that point, attendance behavior creates the network: riders select classes based on personal

preference, repeated encounters occur, and ties form. Once a social tie is achieved, however, the network can begin influencing attendance behavior in return, because established social partners may choose future classes in coordination (think about going to the gym with your gym buddy at a certain time because that works for you both). This creates a feedback loop in which attendance generates social structure and existing social structure subsequently shapes future attendance.

The SoulCycle rider population, along with instructors, studios, and class schedules, were constructed with a heavy reliance on object-oriented programming, with each major entity represented as its own class and assigned attributes relevant to the simulation. Studios stored information such as market, geographic cluster, reservable-bike capacity, and scheduled classes, while instructors were associated with the studios and recurring time slots in which they taught. Riders were modeled as individual objects with their own home market, home cluster, ride rate, and preferences for particular studios, instructors, days, and slots. This structure made it possible for the simulation to preserve heterogeneity across both the physical SoulCycle system and the synthetic rider population rather than treating all riders or studios as interchangeable. The rider population consisted of 10,000 fictitious individuals. Each rider was first assigned to a network market and geographic cluster, which constrained the set of studios they could reasonably attend. These geographic restrictions were especially important in large markets such as Greater New York City, where placing every studio within one unrestricted choice set would allow unrealistic recurring travel patterns between distant locations (for example, despite TriBeCa and Greenwich, CT being in the same network market, I separated them via geographic cluster as it would not be likely for riders to travel between the two studios often, whereas a TriBeCa rider could likely frequent the Financial District location as well given its proximity). Riders were then individually assigned a mean weekly ride rate that determined their general frequency of

attendance, along with a set of preferences that influenced which classes they were most likely to select.

Rider attendance frequency presented one of the more difficult quantities to estimate because SoulCycle does not disclose rider statistic distributions. As a replacement, I used information from SoulCycle’s 2025 “Wrapped” feature. I personally completed 27 classes during the year and was classified by SoulCycle as being in the top 10% of riders, see Figure 2:

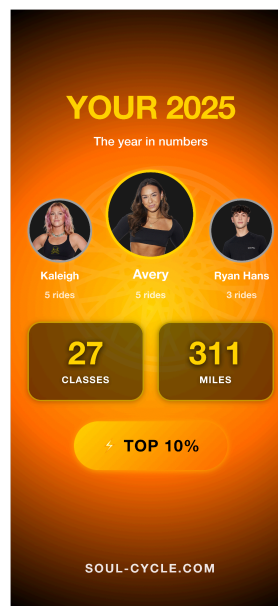


Figure 2: SoulCycle Wrapped Example (SoulCycle 2026)

Rather than interpreting this single observation as representative of the average customer, I used it as evidence of a highly right-skewed attendance distribution: if 27 annual rides is already sufficient to place a customer in the top 10%, this suggests that the majority of riders attend substantially less frequently. I therefore modeled each synthetic rider’s baseline annual ride frequency using a bounded log-normal distribution. A log-normal specification was chosen because it allows a large mass of relatively infrequent riders while retaining a smaller right tail of highly active customers. The 27-ride, top-10% observation served as a calibration reference for the upper portion of this distribution rather than as an exact population percentile constraint.

After calibration, the generated population averaged approximately 11.6 rides per rider per year across simulation seeds, while still preserving substantial heterogeneity in individual attendance frequency, see Figure 3:

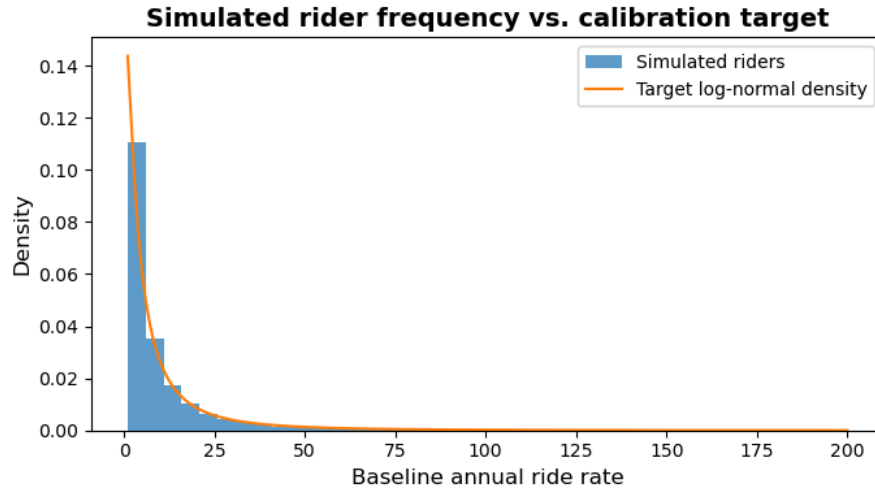


Figure 3: Annual Ride Rate Distribution

Importantly, rider preferences were probabilistic rather than deterministic. A rider with a preferred studio or instructor was therefore more likely to attend classes associated with that option, but was not restricted to doing so exclusively. This engrained recurring behavioral patterns while still allowing enough variation for riders to encounter new people over time. The combination of geographic constraints, heterogeneous attendance frequency, and repeated preferences provides the mechanism through which some pairs of riders repeatedly enter the same classes while most possible rider pairs never interact at all. Class schedules themselves were built from the observed studio and instructor environment described in the data collection stage. Each scheduled class therefore had a studio, instructor, day, slot, and finite bike capacity. During each simulated week, riders made attendance choices from the classes available to them and according to their preferences (restricted to a maximum of one class per day). Once riders were assigned to classes, all rider pairs occupying the same class were recorded as co-attendees,

providing the pairwise interaction data used to update the temporal network. These weekly co-attendance events formed the raw interaction data from which familiarity and social ties were subsequently generated.

To examine whether existing relationships could feed back into future attendance behavior, the model was run under three coordination scenarios. Under no coordination, riders selected classes independently according to their existing geographic and behavioral preferences. Under baseline coordination, riders with established active social ties could coordinate with up to two partners and participate in up to two coordinated classes per week. Under high coordination, these limits increased to four partners and four coordinated classes per week. The scenarios were designed to test whether stronger social influence on attendance would expand the rider network by generating additional interaction or instead concentrate riders within established social routines. Each scenario was simulated for 52 weeks using 10 random seeds, producing 30 total simulation runs. During each run, the model recorded weekly attendance, pairwise co-attendance, familiarity ties, active social ties, and network-level statistics. Final analyses were conducted both longitudinally, to evaluate network growth over the year, and at week 52, to compare the mature network structure across scenarios. Outputs from the simulations were stored at the rider (node), pair (edge), and graph levels, allowing subsequent analysis of individual connectivity, tie progression, component structure, clustering, assortativity, modularity, and other network properties.

Agentic tools were also used throughout the project as a development aid, particularly for running repeated simulation workflows, cleaning and reorganizing/consolidating code, and debugging errors that emerged as the model grew in complexity. These tools helped accelerate repetitive tasks such as executing multi-seed simulation batches, checking outputs, identifying

inconsistencies, and refactoring functions into a cleaner object-oriented structure. Importantly, the underlying modeling assumptions, simulation design, network definitions, and interpretation of results remained researcher-defined; agentic assistance was used primarily to improve implementation efficiency and code reliability rather than to determine the substantive conclusions of the study.

E. Analytical Methods

After constructing the synthetic rider population and simulating weekly attendance, I evaluated the resulting networks using a combination of calibration diagnostics, longitudinal growth measures, node-level statistics, graph-level statistics, and scenario comparisons. Analyses were conducted both across time and at week 52, with results aggregated across the 10 random seeds where appropriate. This allowed the model to be evaluated at multiple scales, from the level of an individual rider's connectivity to the overall fragmentation and community structure of the familiarity network.

The simulated attendance environment seemed to behave consistently across random seeds. Three calibration measures were tracked: the generated mean annual ride rate, seat occupancy rate, and mean weekly class attendance. These measures were compared across all 10 seeds to make sure that stochastic variation did not substantially alter the overall scale of the simulation. I also compared the generated annual rider frequencies to the bounded log-normal distribution discussed in the previous section. Across the calibration runs, the generated mean annual ride rate was approximately 11.6 rides per rider, the average seat occupancy rate was about 69.5%, and mean weekly attendance was approximately 2,221 rider-class attendances. These diagnostics were primarily used to confirm that the model remained relatively stable

across seeds before moving into the actual network analysis. To examine how the rider network developed over time, I summarized both familiarity and active social ties at weeks 13, 26, and 52. At each of these checkpoints, the total number of ties was calculated separately for each seed and then averaged across seeds. Looking at several points throughout the year made it possible to examine whether repeated co-attendance accumulated steadily over time and, more importantly, whether weaker familiarity relationships and stronger active social relationships developed at similar rates.

At week 52, I further evaluated the familiarity network using several whole-network statistics. The number of nontrivial components measured the number of connected rider groups containing at least two riders, excluding single-rider isolates. Largest component size measured the number of riders contained in the largest connected familiarity component, while largest-component share expressed that quantity relative to the full population of 10,000 synthetic riders. Network density measured the proportion of all theoretically possible rider-to-rider connections that were actually present in the network. Since the model contains 10,000 riders, density was expected to be extremely low, so this metric is most informative when interpreted alongside component size and other measures of local connectivity. I also calculated the average clustering coefficient to measure the extent to which a rider's connections were also connected to one another. Degree assortativity was used to determine whether highly connected riders tended to connect to other highly connected riders or instead to riders with fewer connections. Finally, I calculated Louvain modularity on the familiarity network among riders with at least one familiarity tie. This measure was used to evaluate how strongly the network separated into distinct communities rather than behaving as one group. Each of these metrics was calculated separately for every seed and coordination scenario and was summarized using the

mean and standard deviation across the 10 seeds. Network-level metrics were complemented by rider-level measures of connectivity. Familiarity degree, or the number of familiarity ties associated with each rider, was examined using a complementary cumulative distribution function (CCDF). The CCDF shows the share of connected riders with at least k familiarity ties and is particularly useful for showing whether most riders accumulate only a small number of connections while a much smaller group becomes substantially more connected. I also divided riders into low-, moderate-, and high-frequency activity tiers based on their baseline annual ride rates. Mean familiarity degree at week 52 was then compared across these tiers in order to assess whether greater attendance frequency translated into more opportunities for repeated exposure and subsequent network formation.

The main experimental comparison focused on the effect of coordination on familiarity-network breadth. Week-52 familiarity-edge counts were compared across the no-coordination, baseline-coordination, and high-coordination scenarios. Since the same set of random seeds was used across all three scenarios, each seed effectively provided a matched comparison under different levels of social influence on attendance. I examined both seed-level observations and the overall distribution of outcomes across seeds in order to distinguish consistent scenario effects from normal stochastic variation. Related graph-level metrics, including component size, density, clustering, assortativity, and modularity, were also compared across scenarios to determine whether coordination changed broader network structure rather than only the raw number of familiarity ties. Together, these analytical approaches provide a framework for evaluating whether the simulated rider network is stable, how it develops throughout the year, which riders become most connected, how fragmented or

community-oriented the resulting structure becomes, and how that structure changes when existing social ties begin influencing future attendance.

F. Results

After constructing the synthetic rider population and simulating weekly attendance, the resulting networks were evaluated using a combination of calibration diagnostics, longitudinal growth measures, node-level statistics, graph-level statistics, and scenario comparisons. The model validation stage saw consistency across seeds over several key metrics (see Figure 4). The mean annual ride rate for riders was 11.56 (about once per month), with results ranging from 11.33 to 11.78 across seeds. The model successfully approximates that 11.6 class average that we aimed to engineer into the model with the aforementioned bounded log-normal distribution. The mean occupancy rate was 69.5%, meaning that on average 69.5% of bikes are filled in a given class. Results ranged slightly from 68% to 71% across seeds. This occupancy rate is reasonable from observable domain knowledge. In the context of our truncated 10,000 rider simulation, the average weekly attendance across the entire ecosystem was 2220.63, with results ranging from 2174.42 to 2265.83. This finding is more so a result of the restriction on the number of riders rather than an actual reflection on weekly SoulCycle ridership.

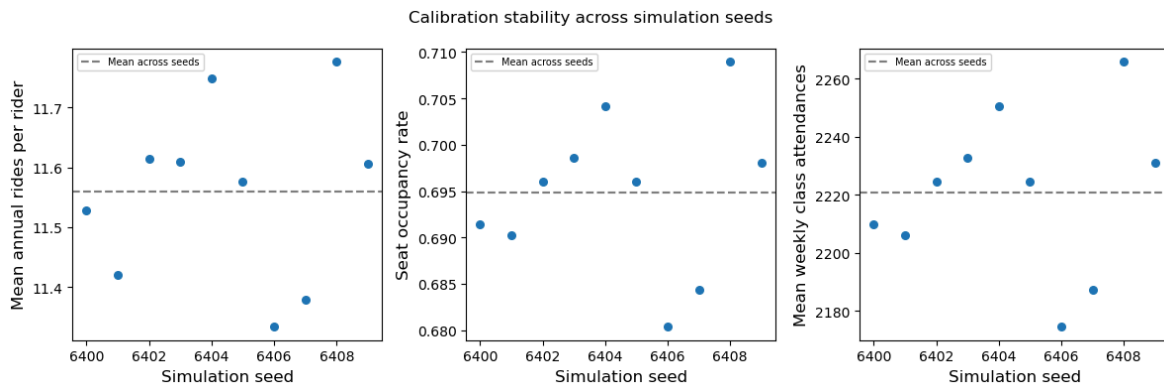


Figure 4: Calibration Metrics

Co-attendances, familiarities, and social tie counts were all updated after each time interval (weekly). Several checkpoints were highlighted to track longitudinal network growth in our baseline scenario, averaging across seeds (see Figure 5). At the 13th week, the simulation, on average, resulted in about 33.6 familiarity ties across the whole network (with a standard deviation of 6.3). Active social ties resulted in an even more stagnant result, with an average of 1.2 social ties (the standard deviation being 1.2). This growth is slow but reasonable given the fact that over 13 weeks (or three months) many riders likely have only gone to three or less classes at that point. With three classes with another rider being the minimum to form a familiarity tie, this initial stagnation is expected. At the 26th week (or about 6 months) the ecosystem has an average of 154.1 familiarity ties (with a standard deviation of 24), and 7 social ties (the standard deviation being 2.5). So familiarity ties have nearly quintupled in another 13 weeks, while social ties are up around 700%, which is reasonable as riders have more classes under their belts, resulting in more opportunities for repetition to turn into recognition or even social connection that extends outside of the studio. At the 52nd week (a full calendar year and the end of our simulation) we find that there are an average of 586 familiarity ties (standard deviation: 58.7) and 36.3 social ties (standard deviation: 8.7).

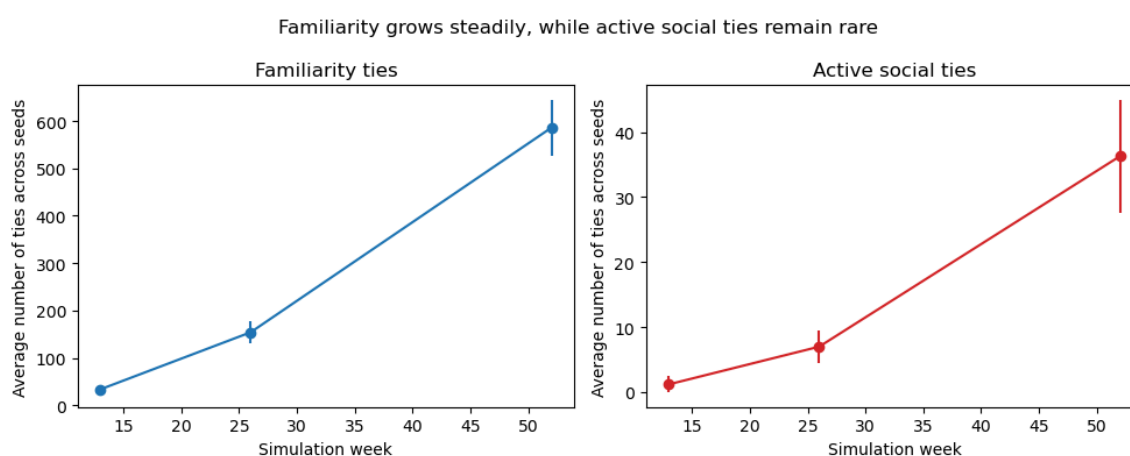


Figure 5: Longitudinal Network Growth

For a network that initialized with 10,000 riders, these results paint a picture of extreme sparsity, which we will corroborate further with density metrics. Furthermore, there is consistency in the struggle to form social ties in this analog environment, even after a full calendar year. This is a major element that will later contribute to latent digital opportunity. Interestingly, while familiarity seemed much easier to form than social ties, as the graphs demonstrate, both growth indicators take on a similarly accelerating shape, begging the question of whether both will continue to take this shape past 52 weeks, and at what point does growth decelerate.

Looking at the network structure at week 52 across all three coordination scenarios further reinforces how sparse and fragmented the simulated ecosystem remains. The number of nontrivial familiarity components, or connected groups containing at least two riders, averaged 45.4 under no coordination, 49.6 under the baseline scenario, and 48.3 under high coordination. The differences are relatively modest compared to the variation across seeds, but baseline coordination produced the highest average number of separate components. At the same time, the largest connected component was actually largest under no coordination, averaging 65.2 riders compared with 55.0 under baseline coordination and 54.4 under high coordination. Even in the most connected case, however, this largest component represented only about 0.65% of the full 10,000-rider population, compared with 0.55% and 0.54% in the baseline and high-coordination conditions. The density results make this sparsity even more apparent. Under no coordination, only about 0.00128% of all theoretically possible rider-to-rider familiarity ties were present, while density fell slightly to 0.00117% under both baseline and high coordination. The clustering coefficient was similarly low across scenarios, averaging 0.69% with no coordination, 0.54% at baseline, and 0.50% under high coordination. Even among riders who do

become connected, closed triangular groups in which a rider's acquaintances are also connected to one another remain relatively uncommon. Interestingly, both density and clustering were highest when riders made attendance decisions independently, providing indications that coordination may actually narrow the breadth of new edge formation by repeatedly concentrating riders around existing social routines.

Degree assortativity was slightly negative in every condition, ranging from -0.172 under no coordination to -0.162 under high coordination. Highly connected riders were somewhat more likely to form ties with riders who had fewer connections than with other highly connected riders. This is consistent with some of the hub-and-spoke structures visible in later network visualizations, where a small number of riders occupy more central positions within otherwise relatively small components. Nonetheless, this is one of the more mild results of this simulation. Finally, modularity (Louvain's method) was extremely high in all three scenarios, ranging from 0.932 to 0.936. These values indicate that familiarity ties were strongly concentrated within distinct communities rather than dispersed evenly throughout the network. The nearly identical modularity scores across scenarios also suggest that changing the level of coordination does not fundamentally change the network's community-oriented structure. Overall, the three conditions produce slightly different levels of breadth and connectivity, but all converge on the same larger finding: the simulated SoulCycle network is extremely sparse, highly fragmented, and strongly divided into localized communities.

Network metric	No coordination	Baseline	High coordination
Nontrivial components	45.4 $\pm$ 5.9	49.6 $\pm$ 9.1	48.3 $\pm$ 6.3
Largest component	65.2 $\pm$ 15.6	55.0 $\pm$ 17.0	54.4 $\pm$ 11.0

Largest-component share	0.65 ± 0.16%	0.55 ± 0.17%	0.54 ± 0.11%
Density	0.00128%	0.00117%	0.00117%
Clustering coefficient	0.69 ± 0.12%	0.54 ± 0.09%	0.50 ± 0.10%
Degree assortativity	-0.172 ± 0.068	-0.164 ± 0.063	-0.162 ± 0.055
Modularity	0.932 ± 0.006	0.936 ± 0.007	0.936 ± 0.006

Figure 6: Network Metrics Table

These aggregate metrics describe the network numerically, but the structure becomes more intuitive when examining the individual connected components themselves:

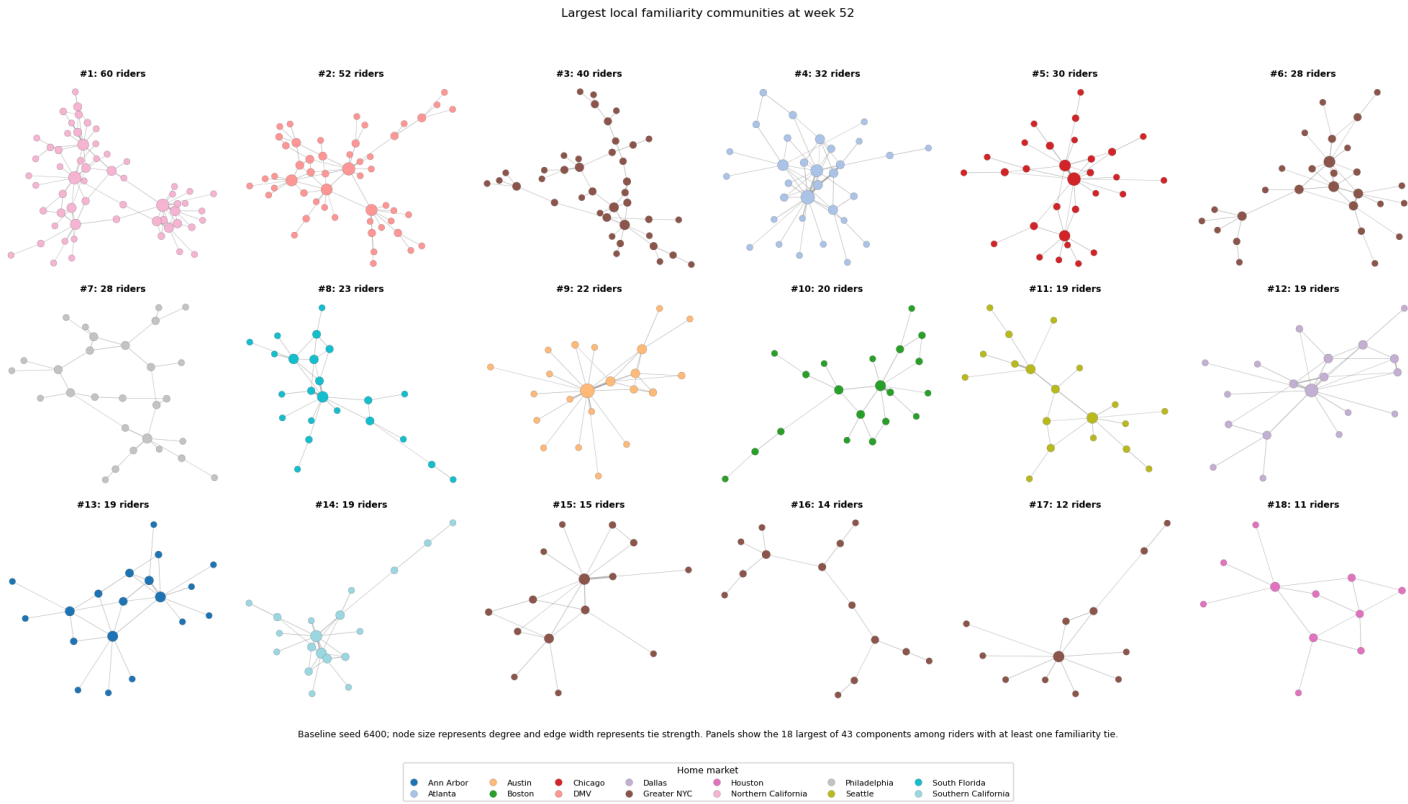


Figure 7: Top 18 Nontrivial Connected Components (6400 Baseline)

The above visualization displays the top 18 of 43 nontrivial connected components in the baseline scenario of our initial “6400” seed. I demonstrate this to show several elements of our ecosystem. First, the market variable essentially locks these subgraphs to be localized within

markets, with no crossover ties formed between markets. Additionally, these networks take on a variety of shapes, but several take on a hub-and-spoke architecture where there are a handful of nodes with high degree and betweenness centrality while some others have greater amounts of equilibrium across node- and edge-level attributes. Despite these differences in internal structure, there is one major similarity across essentially all of the components: they remain small and disconnected from the broader rider ecosystem. Five out of the 18 graphs above fall into the Greater NYC area, demonstrating that even the largest market breaks off into several large subgraphs. The visualization therefore reinforces the same pattern suggested by the component, density, and modularity statistics. Analog familiarity does emerge, but it does so primarily through a collection of localized pockets rather than through one large, integrated SoulCycle network.

The next question was whether this limited connectivity was distributed evenly across riders. Looking at the familiarity degree distribution reveals that it was not. Among riders who formed at least one familiarity tie, most accumulated only a very small number of connections, while a much smaller tail of riders developed considerably higher degrees. This produces a heavily right-skewed degree distribution, suggesting that the network contains a relatively small population of especially well-connected riders rather than distributing familiarity uniformly throughout the rider base, see below:

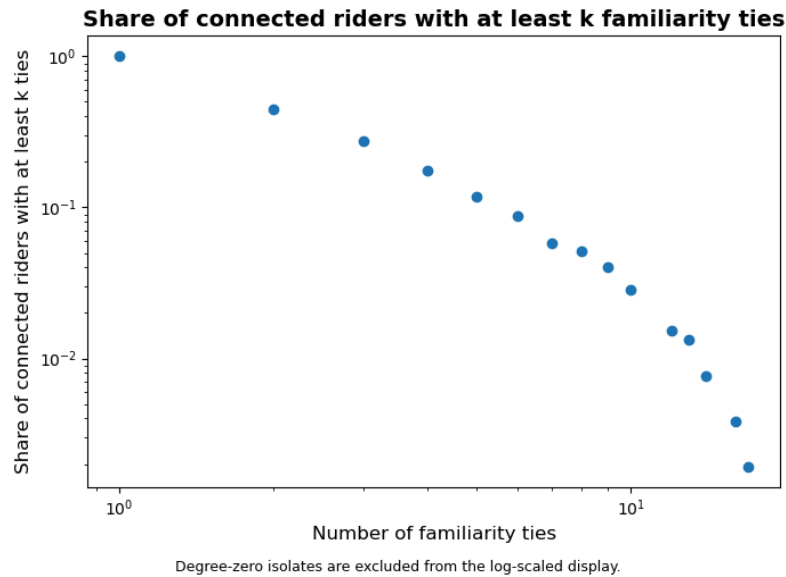


Figure 8: Familiarity Degree Distribution (CCDF)

Attendance frequency provides one intuitive explanation for this imbalance. When riders were separated into low-, moderate-, and high-frequency activity tiers, high-frequency riders had an average familiarity degree of roughly 2.3 by week 52, more than twice that of riders in the low- and moderate-frequency groups, which both remained around one tie per connected rider:

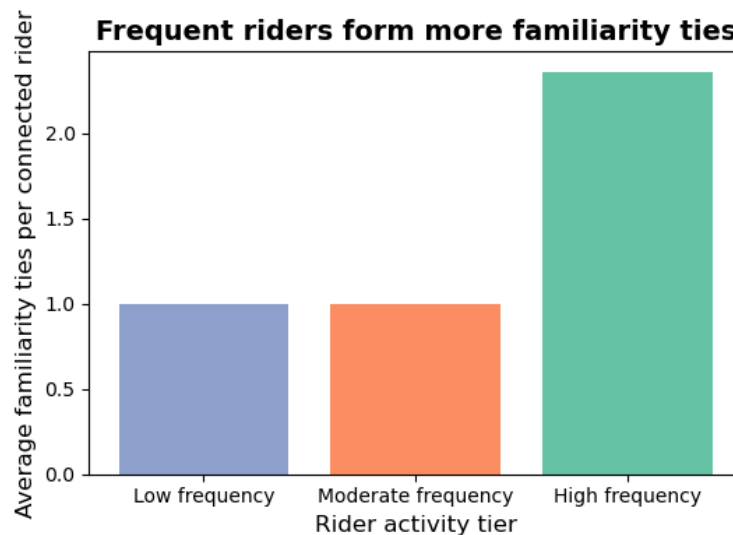


Figure 9: Mean Familiarity by Ride Frequency

This relationship makes sense within the mechanics of the model: riders who attend more frequently simply enter more classes and therefore receive more opportunities to repeatedly encounter the same people. Importantly, this should not necessarily be interpreted as high-frequency riders preferentially seeking one another out. Instead, it primarily reflects unequal exposure opportunities within the network.

Perhaps the most unexpected result came from comparing the three coordination scenarios. It would be reasonable to assume that allowing socially connected riders to coordinate future classes would result in a larger or more interconnected rider network. However, the opposite pattern emerged:

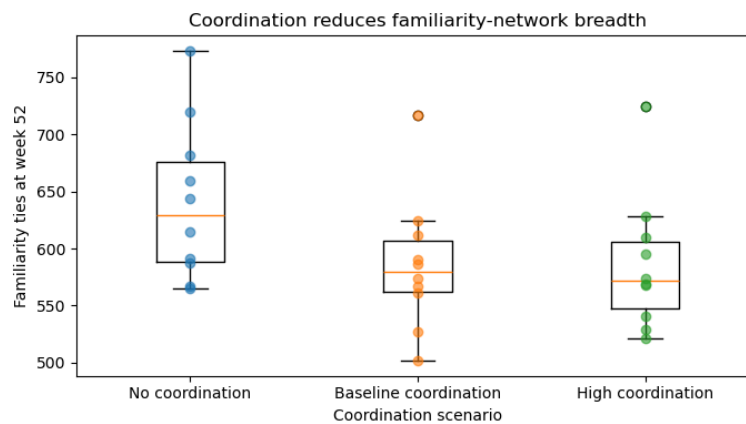


Figure 10: Familiarity Degree Boxplots by Scenario

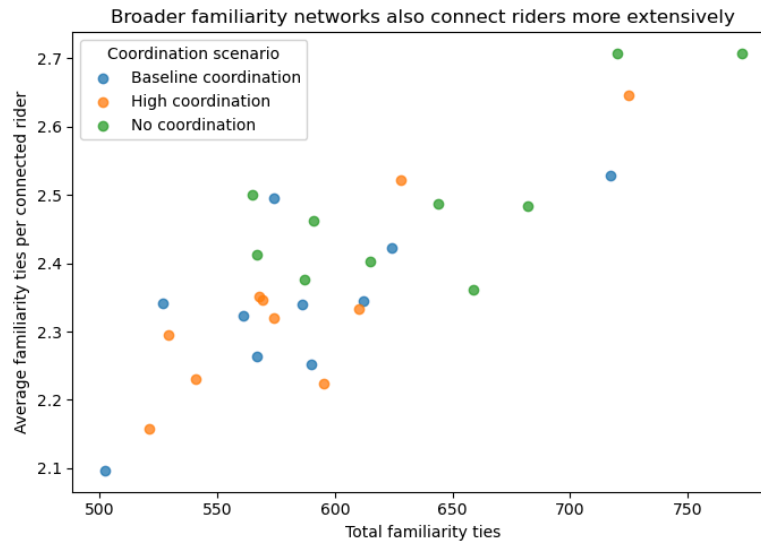


Figure 11: Mean Familiarity Degree Scatterplot by Scenario

By week 52, the no-coordination scenario generally produced the greatest number of familiarity ties, while both baseline and high coordination produced fewer. This same basic pattern was reflected in several of the broader network metrics, with no coordination also producing the largest average component, highest density, and highest clustering coefficient. This creates what I call the “coordination dilemma.” Once riders begin coordinating attendance with people they already know, a portion of their available classes becomes concentrated around existing relationships. While this may reinforce those relationships, it simultaneously reduces the number of opportunities available for encountering new riders and forming additional familiarity ties. In contrast, riders who continue making attendance decisions independently are exposed to a somewhat broader range of co-attendees over time. Coordination therefore appears to strengthen existing routines at the expense of familiarity-network breadth. This does not necessarily mean that coordination produces a socially worse outcome; the model is much better equipped to measure the number and structure of relationships than their subjective quality.

This distinction becomes especially important when examining the progression from simple co-attendance into stronger relationships. By the end of the simulation, thousands of rider pairs had encountered one another repeatedly without ever crossing the threshold into an active social tie. Approximately 2,700 pairs had experienced repeated co-attendance while remaining below the active-social threshold, and roughly 2,200 pairs were only one additional shared encounter away from forming a familiarity tie. By comparison, the number of pairs sitting immediately below the social threshold was much smaller:

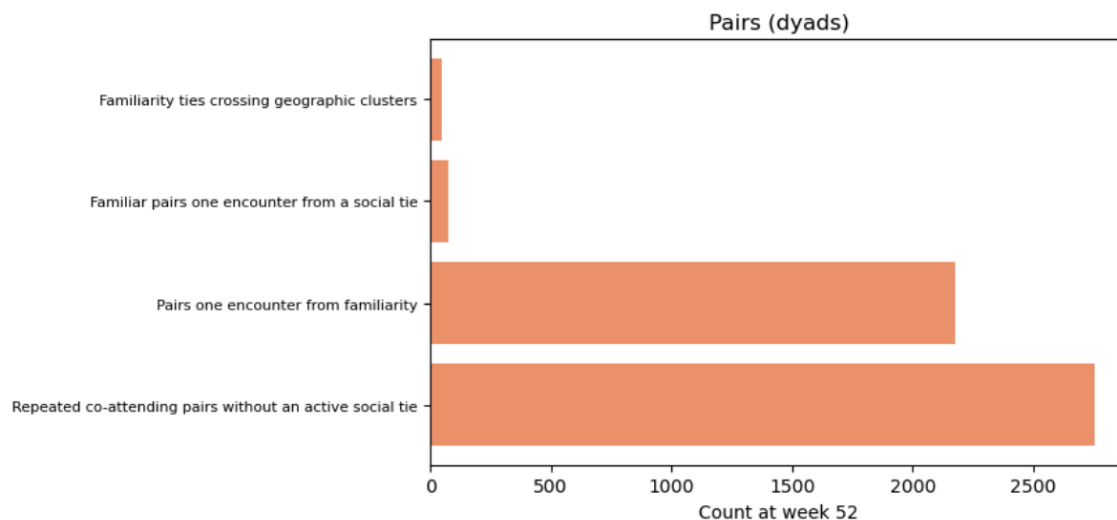


Figure 12: Unrealized Relationships

These results suggest that the largest bottleneck in the simulated network does not occur between complete strangers and familiar riders, but rather between repeated recognition and stronger social connection. Riders frequently develop enough overlap in their schedules to encounter the same people again, but comparatively few dyads accumulate enough interaction to progress all the way through the pipeline. This helps explain why familiarity ties grow substantially faster than active social ties throughout the year and why the active social network remains so small even after 52 weeks. Taken together, the results suggest that repeated analog co-attendance is capable of generating meaningful rider-network structure, but that structure

remains sparse, fragmented, and highly localized. Familiarity develops much more readily than stronger social connection, and riders who attend more frequently receive substantially more opportunities to become connected. At the same time, allowing existing social ties to influence future attendance does not necessarily broaden the network; instead, coordination appears to reinforce already-established routines. These findings provide the basis for the following discussion of what analog community formation can accomplish on its own and where digital infrastructure may be able to supplement it.

G. Discussion

The above results section provides several key characteristics of analog network formation in SoulCycle's specific boutique fitness context. Firstly, analog co-attendance indeed does create a coherent network structure amongst the SoulCycle network. Regular attendance with fellow riders creates a network ecosystem which is sparse, fragmented, and locally bounded. The network market aspect intuitively divides the entire network into subforests of SoulCycle riders, as the feasibility for riders to form strong, reliable ties across the United States is low. Furthermore, within these network markets, a litany of subgraphs, many of which contain fewer than 50 riders. While connected subgraphs take on a variety of structures, several components exhibit hub-and-spoke characteristics where a handful of riders are highly connected and contain high betweenness centrality values. It is also likely that these hub nodes we see across subgraphs may be some of the most active riders, as high ride frequency correlates with higher familiarity degree, on average, than low- and moderate-frequency riders. This actually corroborated the finding in the earlier Prochnow and Patterson article which found in one study that physically active adults have a higher degree, on average, within their health networks than

inactive adults. Encouraging further fitness participation amongst SoulCycle's rider base may produce more active customers and thus more familiarity ties. They can accomplish this by maximizing that aforementioned linking value, but it seems that the analog co-attendance model is limited in the speed and breadth of its growth via linking value.

The most consequential finding is that higher coordination was associated with lower familiarity degree on average. As aforementioned, it is likely that coordination with existing peers as an influence on future ridership likely locks customers into existing social networks, rather than facilitating an expansion of their network. This coordination dilemma is something to consider when weighing reliance on analog versus digital network formation mechanisms, and analog methods may restrict information availability and thus encourage coordination with those you already know. Additionally, we see thousands of repeated co-attendance relationships at the cusp of forming familiarity ties, but likely the typical ridership rate of a SoulCycle customer, along with the decaying tie strength component, likely prevented the repetition necessary to pass the familiarity and/or social thresholds.

My recommendation to SoulCycle is to take a multifaceted approach to solving some of the shortcomings of the analog model. I do not recommend making a formal social network with full ability to make posts, send messages, etc. I think what would be best are a few features to add within their booking/scheduling app to increase exposure to one's local rider base. First, I would add a base level friends feature where fellow riders can follow each other. Once two riders accept each other's friend requests, you then can see on their profiles all their previous classes and all upcoming bookings. Although there will be no messaging feature, riders can link their Instagram, Facebook, etc. in their biography section if riders want to connect on other platforms. The most relevant feature would be the "Recommendations to Follow" feature. This feature will

generate recommendations for riders to follow on the app, guided by similar class preferences, studios, days and time slots. There will also be a comment under each recommendation on how many classes you have already taken with this rider, to perhaps jog your memory on who they are. For example, if you and another rider tend to take classes with Hip-Hop musical themes on Friday mornings, they will be a strong recommendation for you to follow. Furthermore, if you have taken 2 classes with that rider already, it will say so under their recommendation. Amongst your friend list there will also be a running “Best Friends” list when you click on your own profile that will rank the riders in your friends list by the amount of co-attendances you have had with them, along with printing the amount on-screen. These features would help convert repeated analog exposure into actionable familiarity without replacing the in-person environment that gives SoulCycle its linking value. The simulation suggests that many rider pairs repeatedly encounter one another but fail to progress into stronger social ties, so a lightweight digital layer could help riders recognize and coordinate with people they already share routines with. Rather than becoming a conventional social network, SoulCycle’s app could simply make existing patterns of co-attendance more visible.

These recommendations should still be interpreted cautiously. The model does not test whether these features would actually increase friendship, attendance, or retention; it only identifies where the current analog environment appears to leave potential relationships unrealized. Overall, the strongest opportunity is therefore not to replace SoulCycle’s analog model, but to use digital tools to strengthen connections that are already beginning to form offline.

H. Limitations

This project contains several important limitations, many of which stem from the limited availability of rider-level SoulCycle data. The model relies on publicly available schedules, studio locations, instructor assignments, and reservable-bike capacity rather than actual booking histories. As a result, rider behavior, preferences, and social progression had to be simulated rather than directly observed. There is also a temporal limitation, as one observed schedule week from July 20–26, 2026 serves as the basis for the full 52-week simulation. This means that the model does not capture seasonal schedule changes, instructor turnover, cancellations, holidays, or other changes that occur throughout a real calendar year. Additionally, the simulation measures network formation rather than downstream behavioral outcomes. It cannot determine whether stronger social ties would actually increase rider attendance, customer loyalty, or long-term physical activity. The familiarity and social thresholds used throughout the model should similarly be interpreted as operational definitions of increasing relational opportunity rather than direct observations of friendship. Finally, there is no explicit digital mechanism within the simulation through which riders can identify one another, exchange contact information, or form groups. Therefore, while the results may reveal areas of latent opportunity for digital support, they do not demonstrate that any particular digital product or feature would successfully strengthen the network.

I. Conclusion

This project sought to understand how repeated analog co-attendance within SoulCycle could generate rider networks in the absence of a formal digital social platform. Across the simulation, repeated attendance was capable of producing meaningful familiarity structure, but

the resulting network remained extremely sparse, fragmented, and locally bounded. Familiarity ties developed substantially more frequently than stronger social ties, while higher-frequency riders accumulated more connections because they received more opportunities for repeated exposure. One of the most notable findings was the coordination dilemma. Allowing existing social ties to influence future attendance did not broaden the familiarity network; instead, riders became increasingly concentrated within existing social routines. At the same time, thousands of repeated co-attendance relationships remained just below the model's familiarity or social thresholds, suggesting that much of the network's unrealized potential exists between repeated recognition and durable social connection.

SoulCycle therefore does not necessarily need to replace its analog community model with a conventional social network. Rather, lightweight digital features could supplement the linking value already generated inside its studios by helping riders recognize and connect with people they repeatedly encounter. While the present simulation cannot establish whether such features would increase friendship, retention, or physical activity, it demonstrates how analog co-attendance alone can produce social structure and identifies where digital infrastructure may be most useful in extending it.

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